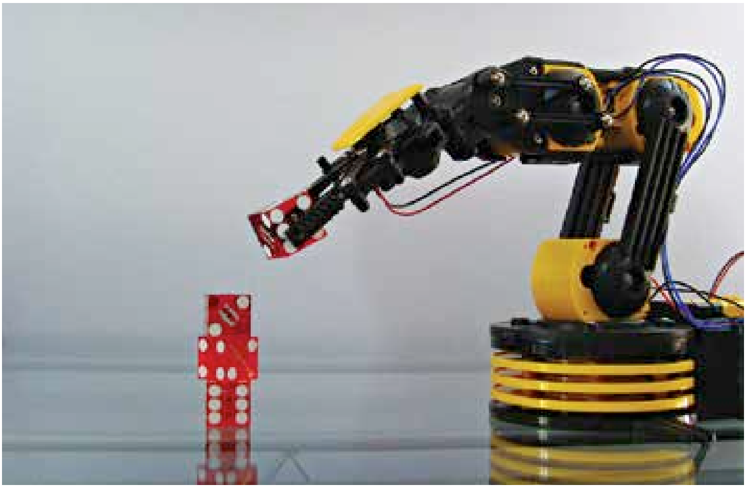

Source: https://c1.staticflickr.com

A simple single degree-of-freedom (the clamp can contract or elongate) actuator on a robotic arm.

motors and stepper motors, which are extremely precisely controllable motor that offers a limited range of rotation with high accuracy. Gear-belt mechanisms are used to change rotations into translations.

Simultaneously, the fields of pneumatics and hydraulics have also arisen, in which high-pressure air, water or oil are used to set object in motion. Pneumatics offer the huge advantage of being variably powered, easily modified, simple, precise and most importantly - safe. In fact, the actuation in several kinds of heavy machinery like bulldozers and tractors is pneumatic.

How it moves - mechanisms in robotics

Robotics is a field full of hidden delights, and the simplest solution is rarely ever the best one. The field of mechanisms has evolved over the past few centuries, starting with the simple study of piston motion, to produce a host of amazingly simple geometric solutions to various robotic problems.

The overall concept is that the constraints between rods of different lengths can be manipulated to generate a host of different curves. Thus, we find the piston's oscillatory mechanism being simply displayed by a three-bar mechanism - the ground, a short crank that turns round and round, and a connecting rod constrained to move in a line in the same plane.